

(a)

collimated plane wave

Focusing principle

finest outer zone

$\Delta r_N \approx 20 \mu\text{m}$

binary-amplitude
zone platelight from open zones
arrives in phase
(half-wave path match)
 $\Rightarrow$ adds constructively

focus F

focal length $f = 150 \text{ mm}$

$r_n \approx \sqrt{n\lambda f}$

$\Delta r_N \approx \frac{\lambda f}{2R}$

$\text{NA} \approx \frac{R}{f}$

 $R = 2 \text{ mm}$

Zone structure (top view)

(b) $R = 2 \text{ mm}$

$\Delta r_N \approx 20 \mu\text{m}$

 r_1 r_3 open zones ($T = 1$, white) / blocked zones ($T = 0$, black), $r_n = \sqrt{n\lambda f}$